Selected-branch convention

Kirtus Justus

1. Notation and normalized instrument branch

Let $\mathcal{B} : M_4 \to M_4$ be a **nonzero** completely positive trace-nonincreasing map. Write $H = \mathcal{B}^*(I_4)$ for its positive success effect, so $0 \le H \le I_4$. Define $p_* = \operatorname{Tr} H/4 > 0$. The trace-one Choi state C of the branch is the Choi operator of $\mathcal{B}$ acting on the output half of the normalized maximally entangled state, divided by p_*. Equivalently, it has Kraus representation

$$C = \sum_j |K_j\rangle\!\rangle\langle\!\langle K_j|, \quad |K\rangle\!\rangle = \tfrac{1}{2}\operatorname{vec} K, \quad \sum_j K_j^\dagger K_j = G := H/p_*, \quad \operatorname{Tr} G = 4.$$

Thus $C \ge 0$ and $\operatorname{Tr} C = 1$; its input-reference marginal is $G^{\mathsf{T}}/4$. A postselected branch is not generally represented by a normalized CPTP Choi state, because G need not be I_4.

All computational basis vectors use the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Let $\mathcal{L}$ be the four-dimensional Hilbert-Schmidt subspace of matrices $V \otimes I_2$. Its normalized Choi-population is $w = \operatorname{Tr}(\Pi_{\mathcal{L}} C)$. Let $g_0 = \langle 00|G|00\rangle > 0$ and $g_1 = \langle 01|G|01\rangle \ge 0$. Let β be the **conditional** probability that the second output qubit is 1 given input $|00\rangle$ and success of this same branch.

2. Theorem: sharp branch-relative flip bound

Theorem 1. For the stated branch, without a trace-preservation or unitality assumption,

$$\boxed{4(1-w) \;\ge\; g_0\beta + \tfrac{1}{2}\left(\sqrt{g_0(1-\beta)} - \sqrt{g_1}\right)^2.} \tag{1}$$

For every triple $g_0 > 0$, $g_1 \ge 0$, $g_0 + g_1 \le 4$, and $\beta \in [0,1]$, equality is realizable by a nonzero CPTNI branch after normalizing its Choi operator.

Proof. Partition each K_j into 2×2 blocks $(A_j, B_j; C_j, D_j)$ indexed by the second bit. Orthogonal projection onto $V \otimes I_2$ replaces its diagonal blocks by $(A_j + D_j)/2$ and removes the others. Therefore

$$4(1-w) = \sum_j \left[\tfrac{1}{2}\|A_j - D_j\|_F^2 + \|B_j\|_F^2 + \|C_j\|_F^2\right].$$

Concatenate the first columns of these blocks over j. Their squared norms obey

$$\|A_0\|^2 = g_0(1-\beta), \quad \|C_0\|^2 = g_0\beta, \quad \|B_0\|^2 = \alpha, \quad \|D_0\|^2 = g_1 - \alpha,$$

with $0 \le \alpha \le g_1$. The reverse triangle inequality gives

$$4(1-w) \ge g_0\beta + \alpha + \tfrac{1}{2}\left(\sqrt{g_0(1-\beta)} - \sqrt{g_1 - \alpha}\right)^2.$$

For fixed g_0, g_1, β, the right side is nondecreasing in α; its interior derivative is $\frac{1}{2} + \frac{\sqrt{g_0(1-\beta)}}{2\sqrt{g_1-\alpha}} \ge 0$. Setting $\alpha = 0$ proves (1), with endpoint cases obtained by continuity.

For attainment, put $u = (4 - g_0 - g_1)/2 \geq 0$ and take normalized Kraus operators

$$K_0 = \mathrm{diag}\left(\sqrt{g_0(1-\beta)}, \sqrt{g_1}, \sqrt{u}, \sqrt{u}\right), \qquad K_1 = \sqrt{g_0\beta}\,|01\rangle\langle 00|.$$

They satisfy $\sum K_j^\dagger K_j = \mathrm{diag}(g_0, g_1, u, u)$, have normalized Choi trace one, and saturate every inequality. Multiplying both Kraus operators by any sufficiently small nonzero common scalar makes their sum-of-effects at most I_4, without changing C, w, g_0, g_1 or conditional β. This gives the promised CPTNI branch. □

CPTP case. If $G = I_4$, then $g_0 = g_1 = 1$ and (1) reduces to the original sharp inequality $4(1-w) \geq \beta + \frac{1}{2}(1 - \sqrt{1-\beta})^2$. More generally, equal relative successes $g_0 = g_1 = 1$ suffice for this formula; full trace preservation is not logically required for this particular inequality.

Inverting a bound. For a separately justified lower population $w \geq w_0$, define $L = 2(g_0 + g_1) - 8(1 - w_0)$. Whenever the data are compatible and $g_0 > 0$, equation (1) yields

$$\sqrt{g_0(1-\beta)} + \sqrt{g_1} \geq \sqrt{\max\{0, L\}}, \quad \beta \leq 1 - \left[\frac{\max\{0, \sqrt{\max\{0, L\}} - \sqrt{g_1}\}}{\sqrt{g_0}}\right]^2. \tag{2}$$

When $g_0 = g_1 = 1$ and the supplied exact model populations give $w_0 = 0.99$, (2) gives $\beta \leq 0.039797974644666\ldots$. This is a conditional **population** bound, not a detector measurement. Independently measured success rates and a common applicable branch are prerequisites to interpreting g_0, g_1 physically.

3. Theorem: sharp Choi-purity interval for one prepared input

Fix an input $|\psi\rangle = |00\rangle$ with nonzero branch success. Define

$$t = \mathrm{Tr}\left[C\,(I_4 \otimes |\psi^*\rangle\langle\psi^*|)\right] = \frac{\langle\psi|H|\psi\rangle}{\mathrm{Tr}\,H}, \qquad 0 < t \leq 1.$$

Here t is the *fraction of the trace-normalized Choi state in the selected input-reference block*, not the raw physical success probability. Let $\sigma = \mathcal{B}(|\psi\rangle\langle\psi|)/\,\mathrm{Tr}\,\mathcal{B}(|\psi\rangle\langle\psi|)$ be the full 4×4 normalized output, $P = \mathrm{Tr}\,\sigma^2$, and $\lambda = \lambda_{\max}(\sigma)$.

Theorem 2. Among all normalized Choi states of nonzero CPTNI operations with these exact t and σ, the complete attainable purity interval is

$$\boxed{t^2 P + \frac{(1-t)^2}{12} \;\leq\; \mathrm{Tr}\,C^2 \;\leq\; t^2 P + (1-t)^2 + 2t(1-t)\lambda.} \tag{3}$$

Proof. Let $P_0 = I_4 \otimes |\psi^*\rangle\langle\psi^*|$ have rank four. In the decomposition $P_0 \oplus (I - P_0)$, write C in blocks $A = t\sigma$, B, and D, with $D \geq 0$ supported on a twelve-dimensional space and $\mathrm{Tr}\,D = 1 - t$. Then

$$\mathrm{Tr}\,C^2 = t^2 P + \mathrm{Tr}\,D^2 + 2\|B\|_2^2.$$

The lower bound follows from $\mathrm{Tr}\,D^2 \geq (1-t)^2/12$ and $\|B\|_2^2 \geq 0$, with equality for $B = 0, D = (1-t)I_{12}/12$. For the upper bound, $\mathrm{Tr}\,D^2 \leq (1-t)^2$. Block positivity gives $B = A^{1/2} K D^{1/2}$ for a contraction K, so $\|B\|_2^2 \leq \|A\|_{\mathrm{op}}\,\mathrm{Tr}\,D = t\lambda(1-t)$.

To attain the upper bound, take a top eigenvector $|u_1\rangle$ of σ and any normalized vector $|v\rangle$ in the complementary twelve-dimensional input-reference block. Set

$$|\chi\rangle = \sqrt{t\lambda}\,|u_1\rangle \otimes |\psi^*\rangle + \sqrt{1-t}\,|v\rangle, \quad C_{\max} = |\chi\rangle\langle\chi| + t\sum_{j=2}^{4} \lambda_j |u_j, \psi^*\rangle\langle u_j, \psi^*|.$$

This positive unit-trace state fixes the desired input block and attains the upper formula. Every positive normalized bipartite Choi operator defines a CP map; multiplying that map by a sufficiently small positive factor makes it trace-nonincreasing while leaving its normalized Choi state and conditional output unchanged. Finally, convexly mixing the two attaining normalized Choi states fixes t, σ and continuously covers all intermediate purities. □

Trace-preserving specialization. CPTP implies $t = 1/4$. Formula (3) becomes the previously supplied

$$\boxed{(P + 3/4)/16 \;\le \operatorname{Tr} C^2 \le (9 + 6\lambda + P)/16.}$$

The supplied model's conditional β bound and first-output spectrum yield the earlier 0.807826398288 outward ceiling when the additional readout premises imply the stated bound on β. Equation (3) is sharp when only t, σ are fixed, and its CPTNI constructions need not be TP for $t = 1/4$; the separate TP attainability construction must be invoked to claim TP sharpness.

Corollary: partial local output information

Let $a \in [1/2, 1]$ be the larger eigenvalue of the first output marginal of σ, and $b = \min\{\beta, 1-\beta\} \in [0, 1/2]$ for a calibrated second-output rank-one effect. Define

$$\begin{aligned} x_* &= \operatorname{clip}\left(\frac{a+b}{2} - \frac{1}{4},\, \max(0, a+b-1),\, \min(a,b)\right), \\ P_{\min}(a,b) &= (a-x_*)^2 + x_*^2 + (1-a-b+x_*)^2 + (b-x_*)^2, \\ m &= \min(1, a+b). \end{aligned}$$

Among all normalized CPTNI Choi states with the given t, a, b, the sharp interval is

$$\boxed{t^2 P_{\min}(a,b) + \frac{(1-t)^2}{12} \;\le \operatorname{Tr} C^2 \le\; t^2[m^2 + (1-m)^2] + (1-t)^2 + 2t(1-t)m.} \tag{4}$$

The endpoints follow by minimizing joint-output purity under the measured local marginals and, separately, attaining $\lambda_{\max}(\sigma) = m$, $\operatorname{Tr}\sigma^2 = m^2 + (1-m)^2$. Explicit two-qubit outputs and the block constructions of Theorem 2 attain both, and continuity fills the interval. At $t = 1/4$ this recovers the numerical and algebraic bounds reported for the CPTP local-output model.

Corollary: pure-branch exclusion does not require trace preservation

A **rank-one Choi state** for any nonzero CP operation has a single effective Kraus operator K. For a pure input with nonzero branch success, its normalized output is a rank-one projector on $K|\psi\rangle$. Hence the two output-qubit marginals must have the same unordered eigenvalues. Consequently the local readings permit a rank-one Choi state only if

$$\boxed{1 - a \le \beta \le a.}$$

Conversely, whenever this inequality holds, a pure joint output with the specified local readings exists by Schmidt decomposition and a rotation of the second Schmidt basis. The single Kraus operator $K = |\varphi\rangle\langle\psi|$ is trace-nonincreasing and yields that output on $|\psi\rangle$. Therefore **this is necessary and sufficient for a rank-one CPTNI explanation when only these local output data are fixed**. The same condition was previously established for a pure unitary-channel

explanation in the CPTP class, but the relevant explanation after postselection need not be unitary. Other joint/Choi observations can rule out rank one even when these local readings alone do not.

A mixed conditional joint output from a known pure input excludes rank-one normalized Choi states of a *single, applicable* CP branch, regardless of global trace preservation. If t is separately known and strictly positive, (3) also supplies a quantitative Choi-purity ceiling. Without a positive lower bound on t, a mixed output alone need not give a uniform numerical gap below purity one.